

Data Structures & Algorithms

BCA — Semester III — 2021 — Answer Sheet

Subject Code: CACS201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. What is data structure? Explain its importance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Write an algorithm to convert infix expression to postfix.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Write a recursive program to generate Fibonacci number up to nth terms.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is insertion sort? Trace and sort the following data using insertion sort: 90, 57, 80, 10, 22, 21, 45, 9, 78

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is hashing? Explain with example the collision resolution method open hashing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write the difference between serial and parallel algorithm with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write a program to implement basic operation in queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is circular linked list? Write a function to delete the node from linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What do you mean by Huffman Algorithm? Explain with example. Construct the B tree of order 5 using following data: 20, 10, 26, 55, 80, 11, 9, 60, 67, 55, 22, 76, 56, 45, 34, 100, 150

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What do you mean by MST? Explain Kruskal's algorithm with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.